1

Week 2

Interactive Mapping — Leaflet

ESS 523c: Environmental Data Science Applications: Water Resources

Interactive Mapping — Core Concepts

ESS 523c: Environmental Data Science Applications: Water Resources

Lecture focus

- What makes a map interactive?
- Tile servers, markers, layers, popups, and controls
- Practical implications for teaching and research in a grad seminar

Leaflet

- Leaflet is a very popular, open-source, JavaScript library for interactive maps
- Used by the New York Times, Washington Post, GitHub, OpenStreetMap, Mapbox and CartoDB

Features

- Interactive panning/zooming
- Layered composition: basemap tiles + vector overlays (markers, polylines, polygons)
- Rich annotations: popups, labels, and custom HTML/CSS

Map Tiles

A tiled web map (slippy map) stitches many tile images streamed over the web to form a seamless, zoomable map.

- Conventions:
  - Tiles are 256 × 256 pixels
  - Zoom level 0 renders the world (domain) in one tile; each zoom doubles tiles in X/Y
    - Each zoom level *doubles* in both dimensions, so a single tile is replaced by 4 tiles when zooming in.
  - ~22 zoom levels are common for web maps
  - Web Mercator (EPSG:3857) is the de-facto tiling projection (lat limits ≈ ±85°)

The OpenStreetMap standard,

- Known as Slippy Map or XYZ, follows these and adds more:
- An X and Y numbering scheme
- PNG images for tiles
- Images are served through a Web server, with a URL like http://…/Z/X/Y.png, where Z is the zoom level, and X and Y identify the tile.

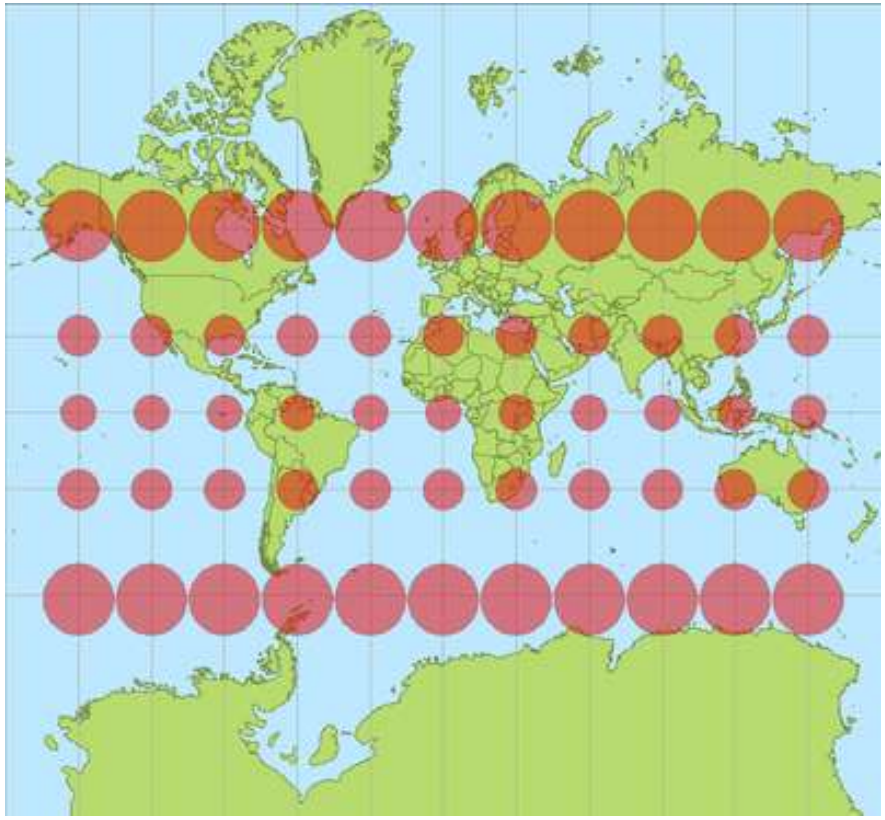
Zoom Levels

- Zoom levels define map scale; higher Z → more tiles and finer detail

Level	# Tiles	Tile width (° of longitudes)	m / pixel (on Equator)	~ Scale (on screen)	Examples of areas to represent
0	1	360	156 412	1:500 million	whole world
1	4	180	78 206	1:250 million	
2	16	90	39 103	1:150 million	subcontinental area
3	64	45	19 551	1:70 million	largest country
4	256	22.5	9 776	1:35 million	
5	1 024	11.25	4 888	1:15 million	large African country
6	4 096	5.625	2 444	1:10 million	large European country
7	16 384	2.813	1 222	1:4 million	small country, US state
8	65 536	1.406	610.984	1:2 million	
9	262 144	0.703	305.492	1:1 million	wide area, large metropolitan area
10	1 048 576	0.352	152.746	1:500 thousand	metropolitan area
11	4 194 304	0.176	76.373	1:250 thousand	city
12	16 777 216	0.088	38.187	1:150 thousand	town, or city district
13	67 108 864	0.044	19.093	1:70 thousand	village, or suburb
14	268 435 456	0.022	9.547	1:35 thousand	
15	1 073 741 824	0.011	4.773	1:15 thousand	small road
16	4 294 967 296	0.005	2.387	1:8 thousand	street
17	17 179 869 184	0.003	1.193	1:4 thousand	block, park, addresses
18	68 719 476 736	0.001	0.596	1:2 thousand	some buildings, trees
19	274 877 906 944	0.0005	0.298	1:1 thousand	local highway and crossing details
20	1 099 511 627 776	0.00025	0.149	1:5 hundred	A mid-sized building

Projection Distortion

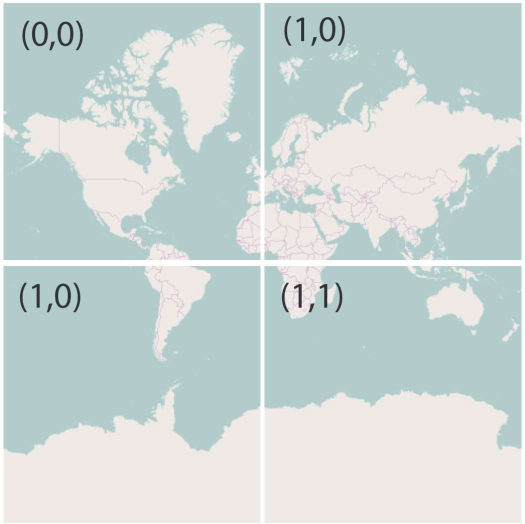
- Web Mercator (EPSG:3857) tiles distort areas, especially near poles; zooming in reveals this.



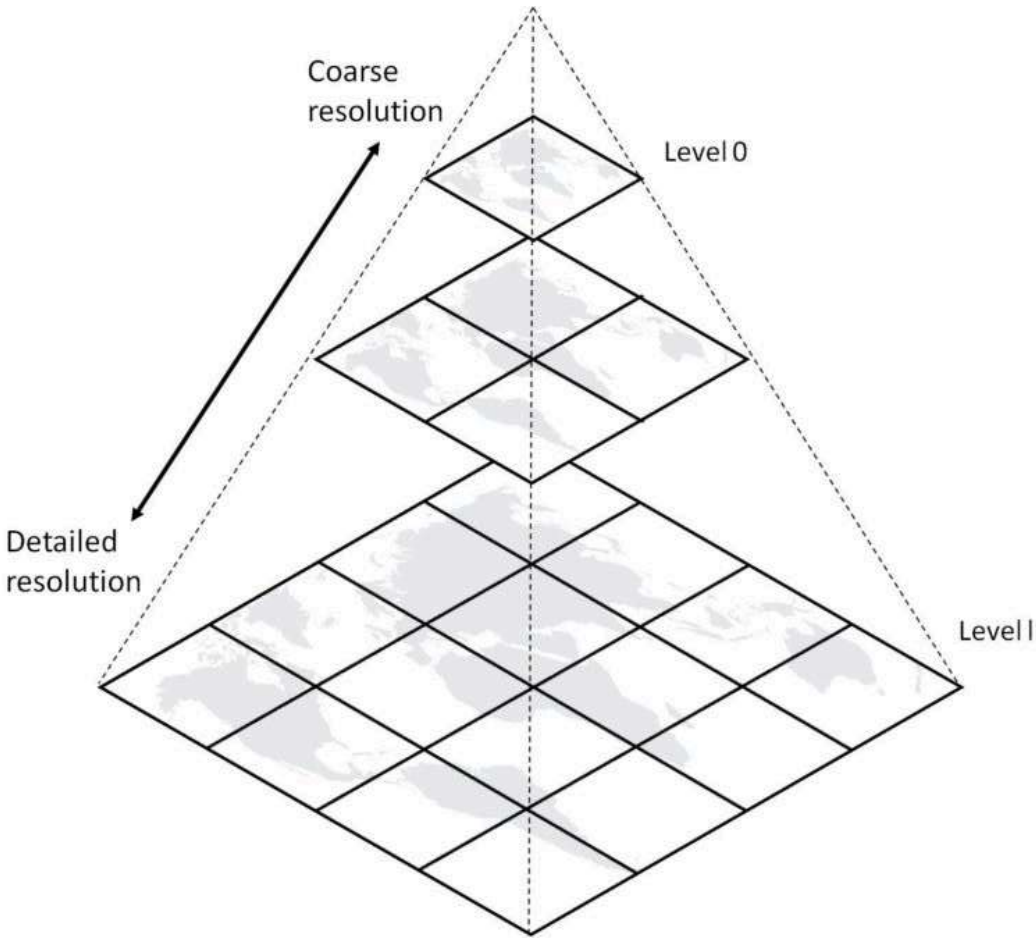
Zoom factor and coordinates

At each zoom the number of tiles increases by 4×; tile coordinates start at (0,0) in the top-left.

At any given zoom level, a specific tile can be identified by cartesian coordinates with 0,0 starting in the top left of the map.



User Zooming



Request Format

Tiles are requested at `url/Z/X/Y.png` — example:

https://a.tile.openstreetmap.org/16/13637/24674.png

The horizontal distance represented by each 256x256 pixel tile, measured along the parallel at a given latitude, is given by: $latitude$ and zoom $zoomlevel$:

$$S_{tile} = C \cdot \cos(latitude)/2^{zoomlevel}$$

where C is the equatorial circumference of the Earth for the reference geoid

$$C = 2pi * 6,378,137.00$$

As tiles are 256-pixels wide, the horizontal distance represented by one pixel is:

$$Spixel = Stile/256$$

For CSU the URL: https://a.tile.openstreetmap.org/16/13637/24674.png

```
C = 2*pi*6378137.00
(Stile = C * cos(40.5699 *pi/180) / 2^16)
#> [1] 464.5005
Stile / 256
#> [1] 1.814455
```

Tile / Zoom — Quick Notes

- **What the formulas do:** C is the equatorial circumference (Web-Mercator reference). $S_{tile} = C \cdot \cos(\mathrm{lat})/2^{\mathrm{zoom}}$ gives the east–west ground width of one 256×256 tile at the given latitude; divide by 256 for meters per pixel.
- **Numeric example:** lat = 40.5699°, zoom = 16 → $S_{tile} \approx 464.5$ m, so one pixel ≈ 1.81 m. At zoom 18 this becomes ≈0.45 m/pixel.
- **Practical tips:** choose the zoom to match desired ground resolution; if a point lies near a tile edge, fetch neighboring tiles (3×3) to avoid cropping; remember this is an east–west estimate along the parallel — Web-Mercator distorts north–south scale.

Live Demos

Live Demos — Examples page

Summary

- Interactive maps are powerful communicative tools for research and teaching
- Key tech: tile providers, vector overlays, popups, controls, and CRS awareness